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Blog

Promising New Solar Cell Tech Uses Mineral Named after Russian Count

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Silicon is the main element used in the creation of photovoltaic solar panels. However, a recent discovery may change that. Perovskite, a mineral named after a 19th century Russian count may lead to cheaper and more efficient photovoltaic solar panels.

It will likely be years before Perovskite could become as commonly used as silicon in PV solar cells, but this new mineral has shown great potential to compete with the more popular material.

Perovskite was named in honor (<http://bit.ly/1EAGLR9>) of the Russian Count Lev Alekseevich Perovski (right) because of his contributions to the development of the Russian mining industry. Count Perovski also maintained an impressive personal mineral collection. This naming honor was given to the Russian Count in 1839, while he was still alive, an honor rarely afforded namesakes of scientific discoveries.



Dramatic Perovskite Improvements

Currently, silicon-based PV solar cells convert light to electricity with around 24 percent efficiency. Prototype perovskite solar cells have already hit 19.3 percent conversion efficiency (<http://bit.ly/1FdRRyv>), with potential to surpass 20 percent. This is exciting news since few materials are capable of exceeding 20 percent conversion efficiency.

Since this emerging solar panel technology is still in the testing phase, it will still take a few years to hit the consumer market. Oxford Photovoltaics intends to release its research on a liquid form of perovskite as well as the method to make solar cells with it. “The company expects products based on its technology to hit the market by 2017,” says (<http://on.wsj.com/1x8UnTR>) Uclia Wang of *The Wall Street Journal*.

Potential for Widespread Adoption

While talk of efficiency and emerging technology is sure to excite fans of solar cells, the real issue pushing widespread adoption is cost. Silicon-based solar cells have dropped in price some 95 percent (<http://bit.ly/18Oqr45>) over the last decade. Can perovskite-based solar cells compete? Martin Green of the University of New South Wales, Australia, thinks so. Green says that solar cells using perovskite “can be made with very simple and potentially very cheap (<http://bit.ly/18Oqr4i>) technology.”

Solar cell efficiency will surely prove the true measure of cost savings, but the upfront cost of the solar cells will likely drive early consumer adoption, which in turn will propel technological innovation. But for those thinking we’re nearing the point where we can just spray any surface with liquid perovskite and start generating electricity, Henry Snaith, a physicist at Oxford University explains that we’re not there just yet: “It is highly unlikely (<http://bit.ly/18Oqr4i>) that anyone will ever be able to just buy a tub of ‘solar paint,’ but all the layers in the solar cell can be fabricated as easily as painting a surface.”

This means the cost of mass production for perovskite solar panels may be cheaper than early silicon panels at the outset. “But historically, it has taken over a decade to scale up novel solar-cell technologies, and a decade from now silicon solar cells could be too far ahead to catch,” says Kevin Bullis at MIT Technology Review.

Recent, Rapid Improvements

To get a better understanding on how quickly this technology is improving, it may help to know the short road this mineral has taken to becoming a major potential competitor. Perovskite was first tested with solar cells in 2009, but MIT Technology Review reports that the conversion efficiency was only 3.5 percent. Compounding the issue, a liquid electrolyte used in the conversion process dissolved the perovskite, which made testing difficult.

It wasn’t until the summer of 2012, when engineers thought to replace the liquid electrolyte with a solid material, that perovskite became stabilized. Since then, conversion has increased steadily.

Takeaway

Competition in solar cell technology will only help drive down costs and speed up technological advancements. This is great news for consumers who may still be on the fence about purchasing PV solar cells for their homes. We may still be a few years out before perovskite solar cells can be a viable option for homeowners. But, the development of a true competitor to silicon-based PV solar cells may be just the thing to create a wide-spread demand for solar panels on the average American home.

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